

High-Precision Epoxy & Eutectic Die Bonder

Authoritative Model Technical Manual & Diagnostic Reference | VAI-MAN-DA-001

NOTICE: SYNTHETIC PROTOTYPE TECHNICAL MANUAL

This document is artificially generated for the VectorAI demonstration and software development.

The specifications, thresholds, service-life values, maintenance intervals, and operating parameters are synthetic and must not be used for real industrial equipment operation or maintenance.

1. Machine Overview & Manufacturing Context

Equipment Model:	High-Precision Epoxy & Eutectic Die Bonder	Document ID:	VAI-MAN-DA-001
Machine Type Key:	die-attach	Cleanroom Bay / Stage:	Bay 2: Die Attach & SMT
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Dual-head high-precision die attach system designed for bonding semiconductor micro-dies onto copper leadframes, organic substrates, and BGA interposers with sub-micron placement accuracy.

Process Flow: A wafer ring expander presents singulated dies. An ejector needle pushes the target die upward, where a vacuum pickup collet retrieves it, performs in-flight vision alignment, and places it onto epoxy-dispensed substrate pads under controlled force and thermal conditions.

2. Major Subsystems & Critical Components

Subsystem / Component	Primary Function	Key Parameters	Degradation Indicators
Vacuum Pick-up Collet	Grips die surface securely without inducing mechanical stress.	Vacuum pressure (-kPa), collet tip wear (μm), contact angle.	Vacuum pressure drop (>-60 kPa), die pickup tilt.
Wafer Ejector Needle Mechanism	Pushes die upward from dicing tape carrier.	Needle tip radius (μm), stroke height (mm), sync timing (ms).	Needle tip bluntness, tape puncture, die cracking.
Epoxy Jet Dispense Valve	Dispenses micro-droplets of conductive/non-conductive epoxy.	Dispense pressure (bar), nozzle temp (°C), dot volume (nL).	Nozzle clogging, dot diameter variance (>10%).
Heated Workholder Platen	Heats substrate to promote epoxy wetting and curing.	Platen temperature (°C), vacuum hold-down (kPa).	Thermal gradient (>3°C across strip), heater burnout.

3. Sensor Telemetry & Threshold Matrix

Threshold boundaries configured for real-time telemetry monitoring and automatic anomaly triggers:

Sensor Name & ID	Unit	Normal Range	Warning Range	Critical Range	Direction
Bond Arm Vibration vibration_arm	mm/s	0.1 – 0.45	0.45 – 0.85	0.85 – 2	Higher is Worse
Collet Vacuum Pressure pressure_vacuum	kPa	-90 – -78	-78 – -60	-60 – 0	Higher is Worse
Heater Block Temp temperature_heater	°C	155 – 175	175 – 195	195 – 250	Higher is Worse

4. Deterministic RUL Model (Weighted Degradation)

Base Useful Life: 1,580 operating hours. Model: $RUL = BaseLife * (1 - (0.50 * wear_vacuum + 0.30 * wear_vib + 0.20 * wear_temp))$

Parameter	Sensor ID	Unit	Weight	Healthy Limit	Critical Limit	Direction
Collet Vacuum	pressure_vacuum	kPa	0.50	-82	-50	HIGHER IS WORSE
Arm Vibration	vibration_arm	mm/s	0.30	0.35	0.9	HIGHER IS WORSE
Heater Temp	temperature_heater	°C	0.20	165	200	HIGHER IS WORSE
TOTAL WEIGHT SUM	-	-	1.00 (100%)	-	-	VERIFIED OK

5. Troubleshooting & Anomaly Symptoms Matrix

Symptom & ID	Severity	Related Sensors	Possible Root Causes	Recommended Corrective Action
Die Placement Tilt / BLT Uneven SYM-DA-001	HIGH	pressure_vacuum, vibration_arm	• Collet vacuum seal leak • Ejector pin bent • Epoxy dot volume mismatch	Flush vacuum filter, replace pick collet, and verify epoxy dispense weight.

6. Authoritative Diagnostic Knowledge Base (Failure Scenarios)

Scenario 1: Collet Vacuum Seal Leakage & Ejector Pin Timing Drift (SCEN-DA-001)	Severity: WARNING
Telemetry Sensor Pattern:	pressure_vacuum > -60 kPa, vibration_arm > 0.6 mm/s
Possible Root Causes:	Collet rubber micro-tear Vacuum line solenoid clogging
Recommended Corrective Action:	Flush vacuum solenoid filter, replace silicone collet, and re-teach ejector needle Z-height.
Verification & Recovery Steps:	Perform vacuum hold test > Run 20-die dummy placement and measure BLT with laser profiler (< 2µm variance)